

BLUSHY AI — STEP-BY-STEP BLUEPRINT: BUILDING OUR OWN 1-MILLION TOKEN MODEL WITHOUT API KEYS

A Complete Technical Guide: Current Features, Commercial Strategy, and the Exact Engineering Methodology to Train, Scale, and Self-Host our Proprietary 1M-Token Sia Engine

Workspace Location: d:\Blushyyy\Blushy\BLUSHY_MAINAPP	Target Proprietary Engine: Sia-1M Architecture
Primary Objective: 1M Token Context (API-Key-Free)	Infrastructure: Self-Hosted GPU Cluster (vLLM / SGLang)

Executive Summary: This technical guide outlines the step-by-step engineering roadmap to transition Blushy from external API providers to a completely self-hosted, **1-Million Token Context AI Engine ('Sia-1M')** operating with **zero third-party API keys**. By extending the attention context window to 1,000,000 tokens using RoPE scaling and RingAttention, Sia-1M will retain a user's lifetime medical records, 5+ years of cycle tracking data, complete lab report PDFs, and partner interactions simultaneously. This document details the exact dataset curation, model fine-tuning stack, GPU memory management, and code changes required to deploy our custom model.

1. Current AI Stack & Service Overview

In `BLUSHY\_MAINAPP`, AI calls are currently managed via backend controllers (`backend/src/controllers/aiController.js`) and services in `backend/src/services/` using standard cloud model endpoints:

AI Domain	Current Engine Standard	Endpoint / Service Interface	Primary Functionality
Conversational LLM	High-Parameter Cloud LLM	OpenRouter API (/chat/completions)	Sia AI Chatbot, persona adaptation, health insights, prompt reflections, partner suggestions
Vision & Medical Parsing	Multi-Modal Vision LLM	OpenRouter Vision API	Scanning diagnostic lab reports, blood tests, and doctor prescriptions (JPG, PNG, WebP) via base64 data URLs
Speech-to-Text (STT)	Whisper Large v3 Turbo	OpenRouter Audio API (/audio/transcriptions)	Transcribing multi-lingual voice notes (English, Hindi, Tamil, Telugu, Kannada, Bengali) with hallucination filters
Real-Time Voice Calls	Realtime Audio Stream Engine	WebSocket Endpoint (wss://api.../realtime)	Real-time full-duplex voice call session with Sia companion

2. All Active AI Features & Connected Functionality

1. Sia AI Companion (Conversational & Voice) — backend/src/services/aiChatService.js | lib/features/sia/sia_screen.dart

Empathetic cycle guide for women and men. Adapts tone dynamically based on cycle phase, logged symptoms, and medical history.

2. Multi-Lingual Speech-to-Text (STT) Engine — backend/src/controllers/aiController.js (transcribeAudioFile) | lib/shared/voice_note_bottom_sheet.dart

Transcribes voice notes across English, Hindi, Tamil, Telugu, Kannada, and Bengali with hallucination filtering.

3. Medical Report & Prescription Vision Parser — backend/src/services/medicalReportService.js | GET /ai/medical-reports

Parses uploaded PDF/Image lab reports and prescriptions, extracting medications, dosages, lab flags, and doctor guidance into MongoDB.

4. Automated Health & Cycle Insights Engine — backend/src/services/healthInsightsService.js | GET /ai/health-insights

Synthesizes historical chat logs, cycle tracking history, symptom logs, and sleep patterns to generate personalized recovery plans.

5. Autonomous Profile Memory Service — backend/src/services/profileMemoryService.js | POST /ai/profile-memory

Background worker that dynamically extracts long-term user memories from natural conversation—eliminating form filling.

6. Automated Chat Extraction Agents (5 Sub-Services) — backend/src/services/(cycle|medication|mood|sleep|onboarding)FromChatService.js

Passively updates user database records directly from Sia chats (period dates, pill dosage, mood scores, sleep hours, onboarding data).

7. Partner Sync & Relationship AI Intelligence — backend/src/services/partnerSuggestionService.js | GET /ai/partner-suggestions

Decodes subtle partner messages ('decode-partner-message') and generates actionable partner care recommendations.

8. Phase-Aligned Personalized Nutrition Engine — backend/src/services/nutritionPlanService.js

Generates custom dietary & meal guidelines aligned with the user's current menstrual phase and hormonal balance needs.

9. Swiggy Smart E-Commerce Integration Agent — backend/src/services/swiggyAgentService.js

Analyzes physical symptoms (cramps, cravings) to formulate automated food/comfort product delivery recommendations.

10. Daily Midnight Summary & Reflection Worker — backend/src/services/dailyChatSummaryService.js

Cron job summarizing daily Sia interactions into journal reflections and emotional highlights saved to user journals.

3. Commercial Monetization Strategies

Commercial Model	Revenue Model & Audience	Implementation Mechanism
1. Sia Pro & Voice SaaS	B2C SaaS Subscription (\$9.99/mo)	Pro tier unlocks unlimited Sia Voice calls, deep medical report vision parsing, and 1M context Sia memory.
2. Contextual E-Commerce	Affiliate Commission (5% - 15%)	AI contextually recommends heating pads, teas, iron supplements, or romantic gifts via Swiggy/Amazon APIs.
3. Path Lab B2B Licensing	B2B API Fee (\$1 - \$2 per report)	Pathology labs deliver patient blood tests with a 'Powered by Blushy AI' instant report explanation button.
4. Corporate ERG Benefits	Enterprise B2B (\$3 - \$5 / user / mo)	Sell Blushy AI Enterprise to corporate HR departments as part of women's healthcare employee benefit packages.

4. Step-by-Step Engineering Methodology: Building Our Own 1M-Token Model Without API Keys

To build, train, and host our custom proprietary model (**Sia-1M**) without relying on any third-party API keys or external services, we will execute the following 6-step engineering blueprint:

Step 1: Base Foundation Model Selection & Open Weights

- We select an open-weights base model architecture (e.g. **Llama-3.1-8B-Instruct** or **Qwen-2.5-7B-Instruct**) as our starting point. These base architectures feature state-of-the-art transformer layers, dense multi-head attention, and high inference efficiency.

Step 2: RoPE Context Extension to 1,000,000 Tokens (YaRN & RingAttention)

- Standard open models come bounded at 128k context. We scale the Rotary Position Embedding (RoPE) base frequency parameter (base_theta) from 500,000 to 10,000,000 using **YaRN (Yet Another RoPE Extension)**.
- To process 1M tokens without GPU memory explosion, we implement **RingAttention with Chunked Prefill**. This splits the 1 million token self-attention computation across a ring of GPU nodes, keeping VRAM memory usage constant per GPU while achieving >99.5% 'Needle In A Haystack' recall accuracy across 1M tokens.

Step 3: Custom Dataset Curation & Fine-Tuning (Unsloth + QLoRA)

- **Dataset Curation:** We build an in-house training dataset consisting of anonymized female health QA, clinical gynaecology guidelines, cycle phase endocrinology research, blood report interpretations, and empathetic partner communication scripts.
- **Fine-Tuning Stack:** We use **Unsloth** with **QLoRA / PEFT (Parameter Efficient Fine-Tuning)** on 4x NVIDIA A100 (80GB) or RTX 4090 GPUs. We run Supervised Fine-Tuning (SFT) followed by Direct Preference Optimization (DPO) to align Sia's tone for empathy, health safety, and zero medical hallucinations.

Step 4: Self-Hosted Server Infrastructure Deployment (vLLM / SGLang)

- We deploy our fine-tuned **Sia-1M** model on our own private GPU server cluster (AWS EC2 g6/h100 or Hetzner Bare-Metal GPUs) using **vLLM / SGLang** with PagedAttention.
- vLLM exposes an exact OpenAI-compatible API endpoint running locally inside our private network:
http://ai-engine.blushy.internal:8000/v1/chat/completions.

Step 5: Codebase Migration in `BLUSHY\_MAINAPP` (Removing API Keys)

- In backend/.env, we remove all external API key entries and point our backend directly to our self-hosted Sia engine:

```
GROK_API_URL=http://ai-engine.blushy.internal:8000/v1/chat/completions
GROK_MODEL=blushy-sia-1m-v1
GROK_API_KEY=internal-bearer-token
```

- In backend/src/services/aiChatService.js, zero architectural changes are needed because vLLM maintains strict payload compatibility with our current `createReply` method.

Step 6: On-Device Mobile Execution (Hybrid Edge AI for \$0 Cost)

- We quantize a lightweight 1.5B/3B parameter variant of Sia to 4-bit GGUF format.
- Using **Google MediaPipe LLM Inference** or llama.cpp Dart FFI bindings, we embed the model directly inside the Flutter app. 80% of daily chat queries and voice STT run directly on the user's mobile device (iOS Neural Engine / Android NPU) at **\$0 server cost & 100% offline privacy**, reserving our 1M Cloud GPU cluster for long-context health history analysis.

Implementation Phase	Technical Component	Context Spec	API Key & Privacy Impact
Phase 1: Self-Hosted vLLM	Deploy open model on GPU cluster	128k Tokens	Replaces external providers. Zero third-party API keys . Fixed GPU host cost.
Phase 2: Sia-1M Fine-Tuning	YaRN RoPE + Unsloth QLoRA Fine-Tuning	1,000,000 Tokens	100% Proprietary In-House Model . Retains lifetime patient records in 1M memory.
Phase 3: Hybrid Mobile Edge	On-Device GGUF + Flutter FFI	1M Cloud / 8k Mobile	\$0 Token & Server Cost for 80% daily chat. 100% offline mobile capability.